CDAC -CSEH

Assignment 2

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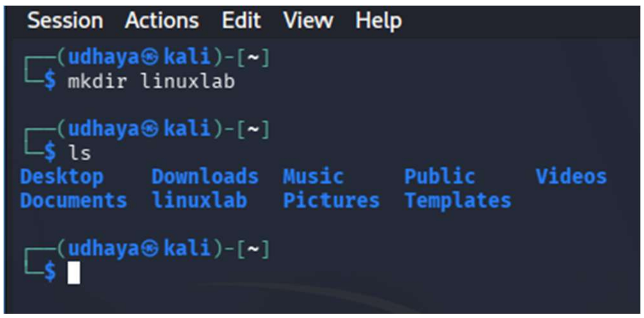
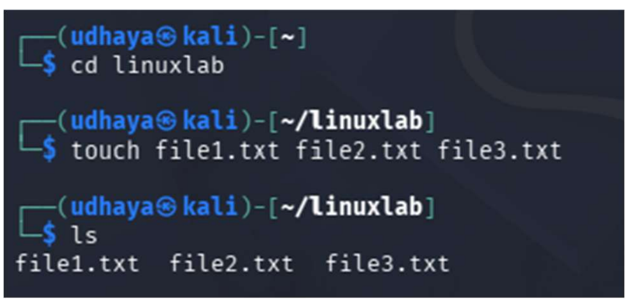
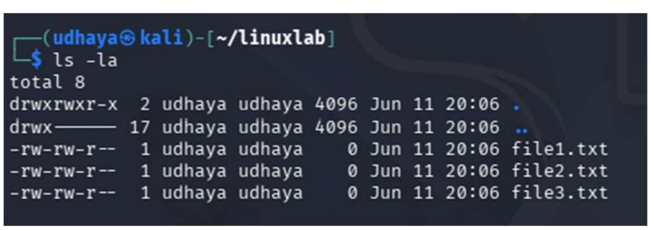
1.Create a directory named `linux_lab` in your home directory and verify its creation.



2.Create three empty files named `file1.txt`, `file2.txt`, and `file3.txt` inside `linux_lab`.



3.Display the current working directory and list all files, including hidden files.



4.Create a hidden file named `.secret` and verify that it exists.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ touch .secret

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -la
total 8
drwxrwxr-x  2 udhaya udhaya 4096 Jun 11 20:07 .
drwx------ 17 udhaya udhaya 4096 Jun 11 20:06 ..
-rw-rw-r--  1 udhaya udhaya    0 Jun 11 20:06 file1.txt
-rw-rw-r--  1 udhaya udhaya    0 Jun 11 20:06 file2.txt
-rw-rw-r--  1 udhaya udhaya    0 Jun 11 20:06 file3.txt
-rw-rw-r--  1 udhaya udhaya    0 Jun 11 20:07 .secret
```

5.Copy `file1.txt` to a new file named `backup_file1.txt`.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ cp file1.txt backup_file1.tx

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls
backup_file1.tx  file1.txt  file2.txt  file3.txt
```

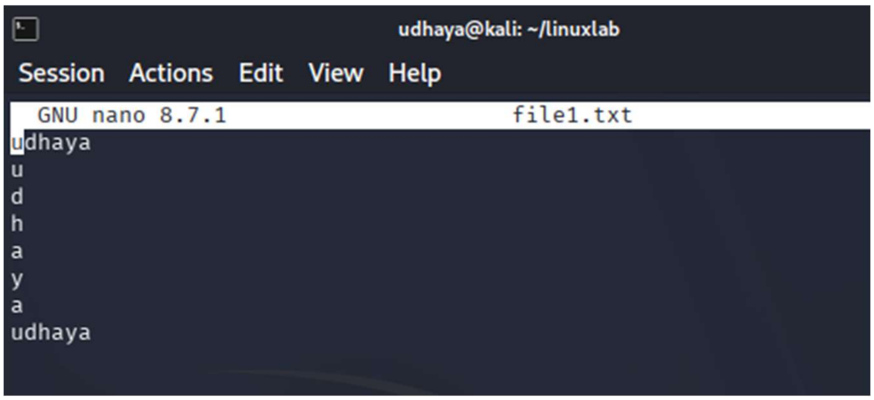
6.Move `file2.txt` to a directory named `archive`.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ mkdir archive

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ mv file2.txt archive/

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls
archive  backup_file1.tx  file1.txt  file3.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls archive
file2.txt
```

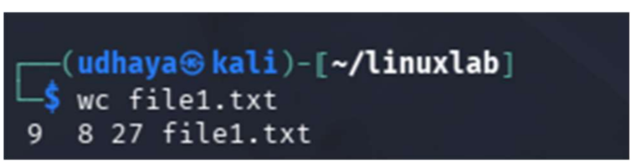
7.Find all files ending with `.txt` inside your home directory.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ find ~ -name "*.txt"
/home/udhaya/linuxlab/archive/file2.txt
/home/udhaya/linuxlab/file3.txt
/home/udhaya/linuxlab/file1.txt
```

8.Display the first 5 lines and last 5 lines of a given text file.



9.Count the number of lines, words, and characters in a file.



10. Search for the word "Linux" in a file and display matching lines.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ grep "Linux" file1.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ grep "udhaya" file1.txt
udhaya
udhaya
```

Process & System Commands

11. Display all running processes and identify the PID of the `bash` shell.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
udhaya      2201    2171  0 20:03 pts/0    00:00:00 /bin/bash
udhaya     11798    2201  0 20:20 pts/0    00:00:00 ps -f

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ps -ef | grep bash
udhaya      2201    2171  0 20:03 pts/0    00:00:00 /bin/bash
udhaya     11985    2201  0 20:20 pts/0    00:00:00 grep --color=auto bash

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ps aux | grep bash
udhaya      2201  0.0  0.2   8484  5548 pts/0    Ss   20:03   0:00 /bin/bash
udhaya     12131  0.0  0.1   6544  2436 pts/0    S+   20:21   0:00 grep --col
or=auto bash
```

12. Start a background process using the `sleep` command for 300 seconds.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sleep 300 &
[1] 12745

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ jobs
[1]+  Running                    sleep 300 &

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ps -ef | grep sleep
udhaya     12745    2201  0 20:22 pts/0    00:00:00 sleep 300
udhaya     12936    2201  0 20:22 pts/0    00:00:00 grep --color=auto sleep
```

13. Terminate the background process started in the previous question.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ kill 12745
bash: kill: (12745) - No such process
[1]+  Done                    sleep 300

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ps -f | grep sleep
udhaya     19916    2201  0 20:36 pts/0    00:00:00 grep --color=auto sleep
```

14. Display system memory usage and disk utilization.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ free -h
               total        used        free      shared  buff/cache   availa
ble
Mem:           1.9Gi       880Mi       223Mi        18Mi       1.0Gi        1.
1Gi
Swap:          1.3Gi          0B       1.3Gi
```

15. Check the IP address and network configuration of your system.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M     0  883M   0% /dev
tmpfs           198M 1004K  197M   1% /run
/dev/sda1        24G   15G  7.1G  68% /
tmpfs           986M  8.0K  986M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           986M   84K  986M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  2.8M  195M   2% /run/user/1000
```

Shell Scripting Basics

16. Write a shell script to display:

- Current Date
- Current User
- Hostname

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ nano q16.sh
```

```
GNU nano 8.7.1                                    q16.sh *
#!/bin/bash

echo "Current Data: $(date)"
echo "Current User: $(whoami)"
echo "Hostname: $(hostname)"
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q16.sh
Current Data: Thursday 11 June 2026 08:45:13 PM IST
Current User: udhaya
Hostname: kali
```

17. Write a shell script to add two numbers entered by the user.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ nano q17.sh
```

```
GNU nano 8.7.1                                    q17.sh *
#!/bin/bash

read -p "Enter First Number: " A
read -p "Enter Sceond Number: " B

SUM = $((A+B))

echo "Sum = $SUM"
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q17.sh
Enter First Number: 12
Enter Sceond Number: 21
Sum = 33
```

18. Write a shell script to determine whether a number is even or odd.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ nano q17.sh
```

```
GNU nano 8.7.1                                                    q18.sh *
#!/bin/bash

read -p "Enter a Number: " N

if [ $((N%2)) -eq 0 ]
then
        echo "$N is Even"
else
        echo "$N is odd"
fi
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q18.sh
Enter a Number: 12
12 is Even

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q18.sh
Enter a Number: 21
21 is odd
```

19. Write a shell script that displays numbers from 1 to 10 using a loop.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ nano q19.sh
```

```
GNU nano 8.7.1                                                    q19.sh *
#!/bin/bash

for i in {1..10}
do
        echo $i
done
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q19.sh
1
2
3
4
5
6
7
8
9
10
```

20. Write a shell script to count the number of files present in a specified directory.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ nano q20.sh
```

```
  GNU nano 8.7.1                                                    q20.sh *
#!/bin/bash

read -p "Enter Directory Path: " DIR

COUNT=$(find "$DIR" -type f | wc -l)

echo "Number of files: $COUNT"
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ bash q20.sh
Enter Directory Path: .
Number of files: 10
```

File Permissions

21. Create a file named `project.txt` and assign the following permissions:

- Owner: Read, Write, Execute

- Group: Read, Execute
- Others: Read only

Display the resulting permission string.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ touch project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ chmod 754 project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -l project.txt
-rwxr-xr-- 1 udhaya udhaya 0 Jun 11 21:01 project.txt
```

22. Change the permissions of `project.txt` using both symbolic and numeric methods to:
`rw-r-----`

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ chmod 640 project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -l project.txt
-rw-r----- 1 udhaya udhaya 0 Jun 11 21:01 project.txt
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ chmod u=rw,g=r,o= project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -l project.txt
-rw-r----- 1 udhaya udhaya 0 Jun 11 21:01 project.txt
```

23. Create a new group named `developers` and assign group ownership of a file to this group.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sudo groupadd developers
[sudo] password for udhaya:

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ cat /etc/group | grep developers
developers:x:1001:

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sudo chgrp developers project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -l project.txt
-rw-r----- 1 udhaya developers 0 Jun 11 21:01 project.txt
```

ACL (Access Control List)

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ which setfacl
/usr/bin/setfacl

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ which getfacl
/usr/bin/getfacl
```

24. Create a user named `student1` and grant read/write access to `project.txt` using ACL without changing the file owner.

Tasks:

- Verify ACL entries.
- Test access using `student1`.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sudo useradd -m student1

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sudo passwd student1
New password:
Retype new password:
passwd: password updated successfully
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ sudo setfacl -m u:student1:rw project.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ getfacl project.txt
# file: project.txt
# owner: udhaya
# group: developers
user::rw-
user:student1:rw-
group::r--
mask::rw-
other::---
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ su - student1
Password:
$ cat /home/udhaya/linuxlab/project.txt
cat: /home/udhaya/linuxlab/project.txt: Permission denied
$ exit

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -ld /home/udhaya
drwx------ 17 udhaya udhaya 4096 Jun 11 20:06 /home/udhaya

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls -ld /home/udhaya/linuxlab
drwxrwxr-x 3 udhaya udhaya 4096 Jun 11 21:01 /home/udhaya/linuxlab

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ chmod o+x /home/udhaya

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ chmod o+x /home/udhaya/linuxlab

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ su - student1
Password:
$ cat /home/udhaya/linuxlab/project.txt
ACL Test Successful
$
```

25. Create a directory named `shared_data` and configure default ACLs so that any new file created inside it automatically gives read/write permissions to user `student1`.

Tasks:

- Verify default ACL settings.
- Create a new file and confirm inherited permissions.

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ mkdir shared_data

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ ls
archive  backup_file1.tx  file1.txt  file3.txt  project.txt  q16.sh  q17.sh  q18.sh  q19.sh  q20.sh  shared_data

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ setfacl -d -m u:student1:rw shared_data

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ getfacl shared_data
# file: shared_data
# owner: udhaya
# group: udhaya
user::rwx
group::rwx
other::r-x
default:user::rwx
default:user:student1:rw-
default:group::rwx
default:mask::rwx
default:other::r-x
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ touch shared_data/testfile.txt

┌──(udhaya㉿kali)-[~/linuxlab]
└─$ getfacl shared_data/testfile.txt
# file: shared_data/testfile.txt
# owner: udhaya
# group: udhaya
user::rw-
user:student1:rw-
group::rwx                    #effective:rw-
mask::rw-
other::r--
```

```
┌──(udhaya㉿kali)-[~/linuxlab]
└─$ su - student1
Password:
$ echo "Hello from student1" >> /home/udhaya/linuxlab/shared_data/testfile.txt
$ cat /home/udhaya/linuxlab/shared_data/testfile.txt
Hello from student1
$ exit
```